



GRADUATION THESIS

ELECTRICAL LOAD FORECASTING USING ELM- PSO NEURAL NETWORK

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CHAPTER 1: INTRODUCTION

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Abstract

This project focuses on short-term electrical load forecasting using a hybrid Extreme Learning Machine and Particle Swarm Optimization (ELM-PSO) model. The study reviews fundamental concepts of load forecasting and analyzes factors affecting electrical load demand. PSO is applied to optimize the hidden-layer parameters of ELM in order to improve forecasting accuracy and model stability. The proposed model is implemented and evaluated using real load data, and its performance is compared with conventional ELM. The results demonstrate the effectiveness of the ELM-PSO approach in enhancing prediction accuracy.

Sinh viên thực hiện
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CHAPTER 1. INTRODUCTION

1.1 Rationale and Objectives

Electricity is a flexible and widely used energy source that plays a vital role in daily life and industrial production. Compared to conventional energy sources such as coal, oil, and natural gas, electricity is more efficient and environmentally friendly. In recent years, the increasing importance of electricity has made power demand forecasting a crucial research topic. Unlike other commodities, electricity cannot be stored in large quantities and must be generated in real time to meet demand [1]. Furthermore, rapid growth in electricity consumption due to population increase and industrial development has created significant challenges for power system operation. Therefore, accurate load forecasting is essential for effective power generation planning.

Electrical load forecasting has been widely studied using various approaches, which can be broadly classified into traditional statistical methods and artificial intelligence-based techniques. Conventional statistical methods such as linear regression and time series models are simple and interpretable, but they often fail to capture the nonlinear and complex characteristics of electrical load demand [2]. In recent years, artificial intelligence methods, including artificial neural networks, support vector machines, and deep learning models, have demonstrated superior performance in modeling nonlinear relationships and handling large-scale datasets. However, many of these methods suffer from high computational cost, slow training speed, and sensitivity to parameter selection. Therefore, there is a growing demand for forecasting models that achieve high accuracy while maintaining computational efficiency and stability.

Extreme Learning Machine (ELM) has attracted significant attention due to its fast-training speed, simple network structure, and strong generalization capability, as it randomly assigns hidden-layer parameters and analytically determines output weights. This characteristic makes ELM particularly suitable for large-scale and real-time load forecasting problems. However, the randomness of hidden-layer weights may lead to unstable prediction performance. Particle Swarm Optimization (PSO), a population-based metaheuristic algorithm, is effective in global optimization with fast convergence and easy implementation [3]. By integrating PSO to optimize the hidden-layer parameters of ELM, the hybrid ELM-PSO model enhances prediction accuracy and robustness while maintaining high computational efficiency.

1.2 Object and Scope

The experimental data used in this study contains hourly records covering the period from **December 2014** to **December 2018**. The dataset includes:

- Total electric load demand (MW),
- Electricity price,
- Temperature data from five Spanish major cities: **Barcelona, Bilbao, Madrid, Seville, and Valencia**,
- Time-related information with hourly resolution.

This dataset reflects realistic short-term load characteristics, such as daily load cycles, strong nonlinearity, and sensitivity to weather conditions. Therefore, it is suitable for evaluating advanced short-term load forecasting models.

The scope of the study is limited to **short-term forecasting with an hourly time horizon**. The research concentrates on **system-level load forecasting**, without considering regional or individual customer-level demand. Input variables are restricted to historical load data, time-related features, and temperature variables provided in the dataset. Other factors, such as economic indicators, population growth, or long-term policy impacts, are not considered.

The study aims to evaluate the effectiveness of a hybrid **Extreme Learning Machine–Particle Swarm Optimization (ELM–PSO)** model in improving forecasting accuracy. The model performance is assessed through offline experiments using standard evaluation metrics, including **MAE, MAPE, and RMSE**, under controlled experimental conditions.

1.3 Methodology

The research methodology adopted in this study follows a **data-driven modeling approach** and consists of several main stages, as illustrated below.

Data Preparation

Historical load and temperature data are collected and examined for missing values and abnormal observations. Data cleaning techniques are applied to ensure consistency and reliability. All input variables are normalized using **Min–Max normalization** to improve numerical stability during model training.

Feature Selection

Input features are selected based on their relevance to short-term load demand. These include:

- Previous load values to capture temporal dependency
- Hourly time information
- Temperature data from multiple locations to represent weather influence

Model Development

A conventional Extreme Learning Machine (ELM) is first constructed as a baseline forecasting model. To overcome the randomness of hidden-layer parameters in ELM, **Particle Swarm Optimization (PSO)** is applied to optimize the input weights and biases of the network.

Model Training and Testing

The dataset is divided into training and testing subsets. PSO is used during the training phase to minimize forecasting error on the training data. The optimized ELM–PSO model is then evaluated on the testing dataset to assess its generalization performance.

Performance Evaluation

The forecasting results are evaluated using **MAE, MAPE, and RMSE**. A comparative analysis is conducted between the conventional ELM and the proposed ELM–PSO model to demonstrate the effectiveness of the hybrid approach.

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Performance Evaluation

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CHAPTER 2. OVERVIEW OF ELECTRICAL LOAD FORECASTING

2.1 Concept and Role of Electrical Load Forecasting

2.1.1 Classification of Forecasting

To provide a systematic overview, electric load forecasting can be classified according to several criteria such as forecasting methodology, level of aggregation, type of load, and forecasting time horizon. These classification approaches and their relationships are illustrated in Figure 1.1.

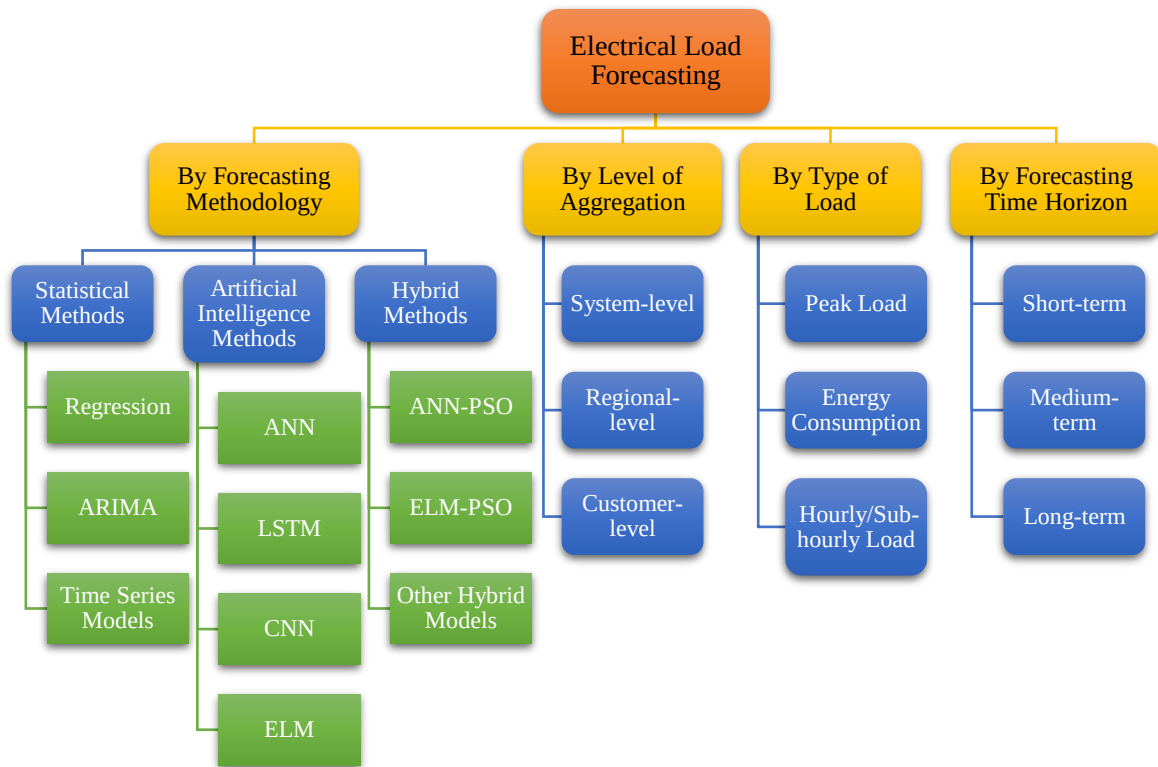


Figure 2.1 Classification of Load Forecasting

In practical applications, the most common and widely accepted classification of electric load forecasting is based on the forecasting time horizon, which includes short-term load forecasting (STLF), medium-term load forecasting (MTLF), and long-term load forecasting (LTLF) [1].

Short-term load forecasting typically covers a time horizon ranging from a few minutes to several days ahead, commonly from 1 hour to 24 or 48 hours. This type of forecasting plays a crucial role in real-time operation of power systems, such as unit commitment, economic dispatch, load balancing, and frequency control. Short-term load demand is highly influenced by weather conditions and time-related factors, making it suitable for advanced data-driven and artificial intelligence-based models.

Medium-term load forecasting focuses on a forecasting horizon from several weeks to months, or up to one year. It is mainly applied in maintenance scheduling, fuel procurement, and operational planning of power generation and transmission systems. Seasonal variations, economic conditions, and consumer behavior are significant factors in medium-term forecasting.

Long-term load forecasting deals with horizons of several years, typically from 3 to 20 years. It is essential for strategic planning, including power system expansion, generation capacity investment, and transmission network development. Long-term forecasting is strongly affected by macroeconomic indicators, population growth, industrial development, and energy policies rather than short-term load fluctuations.

Among these categories, short-term load forecasting is considered the most challenging due to its high volatility and sensitivity to external factors. Therefore, enhancing the accuracy of short-term forecasting remains a critical research topic in modern power systems.

2.1.2 Short-Term Load Forecasting in Power System Operation

Short-term load forecasting (STLF) plays a critical role in the **secure, reliable, and economical operation of power systems**. Accurate short-term forecasts provide essential information for system operators to make timely operational decisions in daily and hourly time scales.

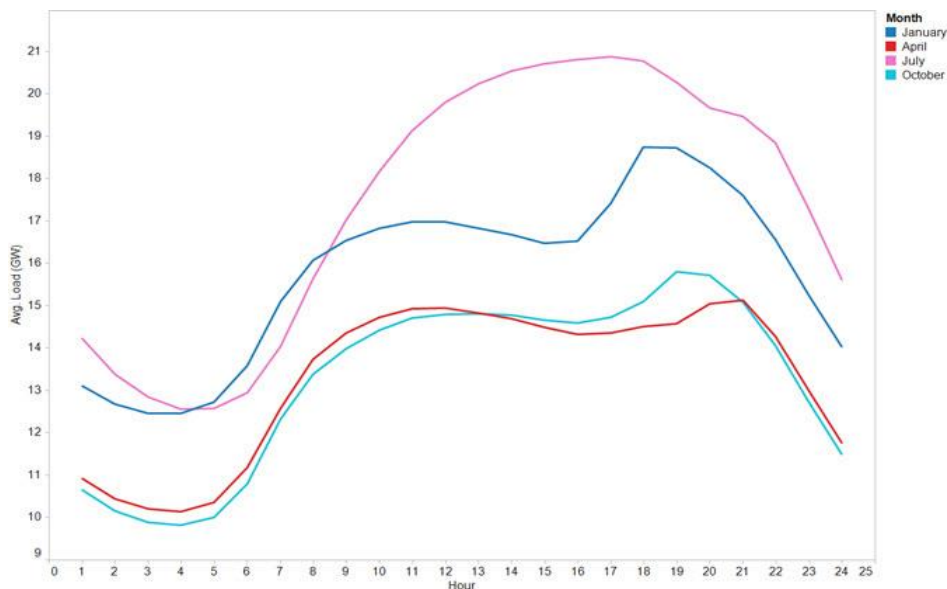


Figure 2.2 Daily demand across seasons (Source: New England ISO)¹

One of the most important applications of STLF is **unit commitment and economic dispatch** [4]. Based on short-term load forecasts, power generation units are scheduled to operate in an optimal manner, ensuring that electricity demand is met at minimum operational cost while satisfying technical constraints. Inaccurate

¹ Available: <https://www.iso-ne.com/isoexpress/web/reports/load-and-demand>. Accessed: Jan. 2026.

forecasts may lead to either excessive reserve allocation or insufficient generation capacity, both of which increase operational costs and system risks.

In addition, STLF is crucial for **real-time balancing and frequency control** in power systems. Since electricity cannot be stored economically on a large scale, generation must continuously match demand. Precise short-term forecasts help system operators maintain power balance, reduce frequency deviations, and enhance overall system stability.

Short-term load forecasting also supports the **integration of renewable energy sources**, such as wind and solar power. The variability and uncertainty of renewable generation increase the complexity of system operation. Accurate STLF enables better coordination between conventional generators and renewable sources, thereby improving system flexibility and reducing the need for expensive backup reserves.

Furthermore, STLF contributes to **preventing system congestion and overload conditions**. By anticipating peak load periods, operators can implement preventive measures such as load shifting, demand response, or network reconfiguration. This helps improve the reliability of transmission and distribution systems and reduces the probability of blackouts.

Due to its strong dependence on weather conditions, time-related factors, and human activities, short-term load demand exhibits high nonlinearity and volatility. Consequently, improving the accuracy of short-term load forecasting remains a major challenge and an active research topic. This has motivated the application of advanced artificial intelligence and hybrid optimization techniques, such as the ELM-PSO model proposed in this study.

2.2 Factors Affecting Electrical Load

Electric load demand is influenced by various factors related to **weather conditions, time characteristics, and socio-economic activities**. These factors interact in a complex and nonlinear manner, making load forecasting a challenging task, especially in short-term horizons. Understanding the impact of these factors is essential for selecting appropriate input variables and improving forecasting accuracy.

2.2.1 Weather Factors

Weather conditions are among the most significant factors affecting electric load demand, particularly in short-term load forecasting. **Temperature** has a direct impact on electricity consumption due to the widespread use of heating, ventilation, and air conditioning (HVAC) systems [5]. Extreme temperatures, either high or low, often lead to sharp increases in electricity demand.

In addition to temperature, **humidity** also influences load demand by affecting human comfort levels and the operational efficiency of cooling systems. Other

meteorological variables such as wind speed, solar radiation, and rainfall may indirectly affect electricity consumption, especially in regions with high penetration of renewable energy sources. Due to their strong correlation with load demand, weather-related variables are commonly used as input features in modern forecasting models.

2.2.2 Time Factors

Electric load demand exhibits clear **temporal** patterns associated with human activities and social habits. Time-related factors include the **time of the day**, **day of the week**, and **seasonal variations**. Typically, daily load profiles show peak demand during working hours and lower consumption during nighttime.

Weekly patterns also exist, with load demand on weekdays generally higher than that on weekends. In addition, special days such as **public holidays** and **festivals** often lead to abnormal load behavior compared to regular days. Incorporating time indicators into forecasting models helps capture periodic patterns and improves prediction performance.

2.2.3 Socio-Economic Factors

Socio-economic factors play a more dominant role in medium-term and long-term forecasting but can also influence short-term demand. Factors such as **population growth**, **industrial activity**, **income level**, and **lifestyle changes** contribute to variations in electricity consumption.

Urbanization and industrial development typically lead to sustained increases in load demand, while economic downturns may cause demand reduction. Government policies related to energy efficiency and demand-side management can also alter consumption patterns. Although these factors change slowly over time, they are essential for understanding long-term load trends and system planning.

2.3 Overview of Existing Forecasting Methods

Due to the complex, nonlinear, and time-varying characteristics of electric load demand, numerous forecasting methods have been proposed and developed over the years. These methods can generally be classified into **traditional statistical methods** and **artificial intelligence-based methods**, each with its own advantages and limitations.

2.3.1 Traditional Statistical Forecasting Methods

Traditional statistical methods are among the earliest approaches used for electric load forecasting. These methods rely on historical load data and mathematical models to identify linear relationships and temporal patterns.

Common statistical techniques include **linear regression**, **autoregressive (AR)** models, **autoregressive moving average (ARMA)**, and **autoregressive**

integrated moving average (ARIMA) models. Regression-based models are simple to implement and provide clear interpretability, but they often struggle to capture nonlinear relationships in demand.

Time series models such as ARIMA are effective in modeling stationary load patterns and short-term trends [4]. However, their performance degrades when dealing with highly volatile data, abrupt load changes, and external influencing factors such as weather conditions. As a result, traditional statistical methods are often insufficient for modern short-term forecasting problems [6].

2.3.2 Artificial Intelligence–Based Forecasting Methods

With the advancement of computational power and data availability, artificial intelligence (AI) techniques have been widely applied to electric load forecasting. These methods are capable of modeling **nonlinear and complex relationships** between input variables and load demand.

Artificial neural networks (ANNs) are among the most commonly used AI methods in load forecasting. Variants such as **backpropagation neural networks (BPNN)**, **recurrent neural networks (RNN)**, and **long short-term memory (LSTM)** networks have demonstrated superior performance compared to traditional methods, especially in short-term forecasting.

In addition, deep learning models such as **convolutional neural networks (CNN)** have been applied to extract temporal and spatial features from load data. Despite their high accuracy, these models often require large datasets, long training times, and careful parameter tuning.

The **Extreme Learning Machine (ELM)**, a single-hidden layer feedforward neural network, has gained attention due to its extremely fast training speed and simple structure. ELM randomly assigns input weights and biases and computes output weights analytically, making it suitable for real-time forecasting applications. However, the randomness of hidden-layer parameters may lead to unstable performance and suboptimal accuracy [7].

Table 2.1 Comparison of Load Forecasting Methods

Energy Load Forecasting Techniques	Specific Model Used	Strength	Weakness
Regression [8]	Linear Regression and Multiple Linear Regression	Very useful in non-real time forecasting. Functional relationship between previous, forecast load and other factors such as weather, time of the day.	Not accurate for real time load and unable to handle nonlinear load consumption. Adding parameters make it unstable.

Time-series Analysis [4]	Auto Regressive Moving Average (ARMA), Auto Regressive Integrated Moving Average (ARIMA), Deterministic decomposition.	They possess abilities to accommodate seasonal component effects.	They suffer from numerical instability.
Artificial Neural Network	Multilayer Perceptrons, Back Propagation Algorithm, Steepest descent Error Back Propagation.	Ability to handle nonlinear relationships in load consumption by adjusting its weight during the training process.	Large amounts of data are needed to train the model and complexity in the training of such data.
Fuzzy Inference System	Defuzzification Method using Centre of Area, Middle of Maxima, Last of Maxima and Centre of gravity	Faster and more accurate in performance including simplicity in rule formation.	Selection of membership functions to form its rule is based on trial and error.
Support Vector Machine	Support Vector Regression using Incremental Learning Algorithm, Support Vector Regression.	It enhances higher feature space dimensionality by using ϵ -insensitive loss for linear regression computation and reduction in model complexity.	Choosing of suitable kernel and difficulties in its interpretation are major concerns.

2.3.3 Evaluation and Selection of the ELM-PSO Approach

To overcome the limitations of individual forecasting methods, **hybrid models** combining neural networks and optimization algorithms have been proposed. These hybrid approaches aim to improve forecasting accuracy by optimizing model parameters and enhancing generalization capability.

Particle Swarm Optimization (PSO) is a population-based optimization algorithm inspired by the social behavior of bird flocks and fish schools. PSO has been widely used to optimize neural network parameters due to its simplicity, fast convergence, and strong global search capability.

By integrating PSO with ELM, the randomness of ELM's hidden-layer weights and biases can be effectively optimized. The resulting **ELM-PSO model** retains the fast-learning advantage of ELM while achieving higher forecasting accuracy and improved stability. Therefore, the ELM-PSO approach is selected in this study as a suitable solution for short-term electric load forecasting.

2.4 Related Works

The field of short-term electrical load forecasting has witnessed a significant transition from traditional statistical methods to Artificial Intelligence (AI) and

Machine Learning (ML) techniques. Recent studies have focused on improving forecasting accuracy and computational efficiency by comparing various architectures and developing hybrid models.

2.4.1 Application of Artificial Neural Networks (ANN)

In the early stages of AI adoption for load forecasting, Feed-forward Neural Networks utilizing backpropagation algorithms demonstrated superior performance compared to classical extrapolation methods.

A notable study by Dang Quang Khoa (2017) [9] applied Artificial Neural Networks to forecast electricity demand for Vinh City, Vietnam, for the period 2016–2020. The research constructed a two-layer feed-forward network with four input variables: population, GDP, industrial production value, and electricity price. The study concluded that ANN models could capture nonlinear relationships better than traditional elasticity coefficient methods. However, conventional ANNs often suffer from slow convergence rates and susceptibility to local minimal due to the gradient-based learning mechanism.

2.4.2 Performance of Extreme Learning Machine (ELM) vs Deep Learning

To overcome the limitations of iterative training in traditional neural networks, the Extreme Learning Machine (ELM) was introduced as a randomized algorithm for Single-Hidden Layer Feedforward Networks (SLFNs).

A recent comprehensive study by Nguyen Thi Hoai Thu et al. (2024) [10] at Hanoi University of Science and Technology evaluated the performance of ELM against state-of-the-art Deep Learning models, including Convolutional Neural Networks (CNN), Long Short-Term Memory (LSTM), and Gated Recurrent Units (GRU), using 2018 load data from Hanoi. The empirical results demonstrated the superiority of ELM in both accuracy and speed:

- **Accuracy:** For 1-step forecasting, the ELM model achieved the lowest Root Mean Square Error (RMSE) of **91.0 MW**, significantly outperforming LSTM (134.1 MW), ANN (136.7 MW), and GRU (146.8 MW).
- **Computational Efficiency:** ELM exhibited an extremely fast training time of approximately **0.48 seconds**, whereas LSTM and GRU required over 4 and 7 minutes, respectively.

This study confirmed that for short-term forecasting tasks, ELM provides a more efficient solution than complex deep learning architectures while maintaining high accuracy.

2.4.3 Research Gap and Proposed Hybrid Approach

While ELM offers significant advantages in speed, its performance relies heavily on the random initialization of input weights and hidden layer biases. This randomness can lead to fluctuations in forecasting stability. In conclusion, Nguyen Thi Hoai Thu et al. suggested that future improvements could be achieved by combining the model with optimization techniques or creating hybrid models.

Addressing this gap, this thesis proposes a hybrid **ELM-PSO** model. By integrating Particle Swarm Optimization (PSO) to optimize the input weights and biases of the ELM, the proposed approach aims to retain the fast-learning capability of ELM while mitigating its inherent instability, ensuring robust and accurate short-term load forecasting.

CHAPTER 3. THEORETICAL BACKGROUND

3.1 Feedforward Neural Networks

A feedforward neural network (FNN) is one of the most fundamental and widely used architectures in artificial neural networks. In an FNN, information flows strictly in one direction, from the input layer through one or more hidden layers to the output layer, without forming any feedback or recurrent connections. Due to its simplicity and strong nonlinear approximation capability, the feedforward neural network has been extensively applied in pattern recognition, regression analysis, and electric load forecasting.

A typical feedforward neural network consists of three main components: **the input layer**, **the hidden layer(s)**, and **the output layer**. The input layer receives external signals, such as historical load data, weather variables, and time-related features. The hidden layer is responsible for extracting nonlinear relationships from the input data through weighted connections and activation functions. The output layer produces the final forecasting result.

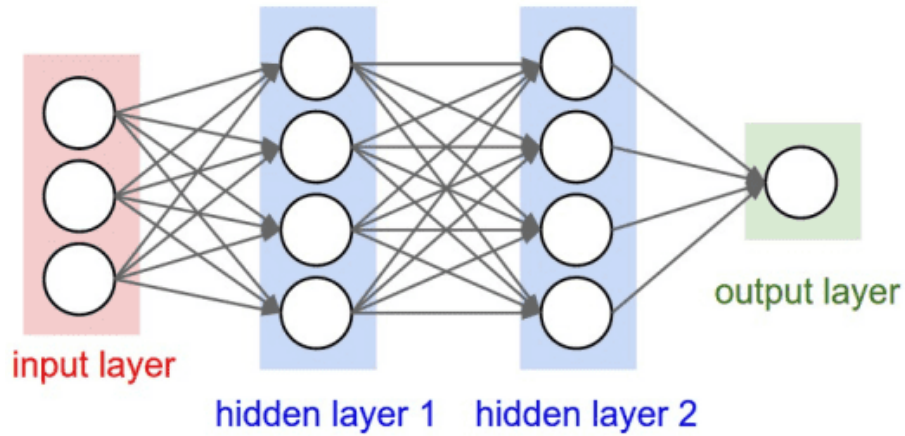


Figure 3.1 An example of a Feedforward Neural Network with two hidden layers
Mathematically, the output of a single hidden-layer feedforward neural network can be expressed as:

$$y = \sum_{i=1}^L \beta_i g(\mathbf{w}_i \cdot \mathbf{x} + b_i) \quad (3.1)$$

where

$\mathbf{x}$ is the input vector,

L is the number of hidden neurons,

$\mathbf{w}_i$ and b_i are the weight vector and bias of the i -th hidden neuron,

$g(\cdot)$ is the activation function, and

β_i denotes the output weight connecting the hidden layer to the output layer.

Common activation functions used in feedforward neural networks include the **sigmoid**, **tanh**, and **rectified linear unit (ReLU)** functions. These nonlinear activation functions enable the network to approximate complex nonlinear mappings between inputs and outputs.

Traditionally, feedforward neural networks are trained using **gradient-based learning algorithms**, such as the backpropagation (BP) algorithm. During training, network weights and biases are iteratively updated to minimize a predefined error function, typically the mean squared error (MSE). Although backpropagation-based FNNs can achieve high forecasting accuracy, they suffer from several limitations, including slow convergence speed, sensitivity to initial parameter values, and the risk of getting trapped in local minima.

In the context of electric load forecasting, these limitations become more pronounced due to the large size of datasets and the highly nonlinear and time-varying nature of load demand. Consequently, alternative learning mechanisms that can improve training efficiency and robustness have attracted increasing research interest. One notable approach is the **Extreme Learning Machine (ELM)**, which is a special form of single-hidden layer feedforward neural network with a significantly simplified training process. The theoretical foundation of ELM is discussed in the following section.

3.2 Extreme Learning Machine (ELM)

The Extreme Learning Machine (ELM) [11] is a learning algorithm developed for **single-hidden-layer feedforward neural networks (SLFNs)**, aiming to overcome the main limitations of conventional backpropagation-based training methods. Unlike traditional neural networks, ELM randomly assigns the input weights and hidden-layer biases and determines the output weights analytically in a single step [11]. This characteristic significantly reduces training time while maintaining good generalization performance.

3.2.1 Architecture of Extreme Learning Machine

The structure of an ELM is similar to that of a single-hidden-layer feedforward neural network, consisting of an input layer, one hidden layer, and an output layer. However, the key difference lies in the training mechanism. In ELM, the **input weights and hidden-layer biases are randomly generated and fixed**, while only the output weights are calculated during the learning process.

Let $\mathbf{x}_j \in \mathbb{R}^n$ be the input vector of the j -th sample, and $y_j \in \mathbb{R}$ be the corresponding target output. The output of the ELM model can be expressed as:

$$\sum_{i=1}^L \beta_i g(\mathbf{w}_i \cdot \mathbf{x}_j + b_i) = y_j, \quad j = 1, 2, \dots, N \quad (3.2)$$

where

L is the number of hidden neurons,

$\mathbf{w}_i$ and b_i are the randomly assigned input weight vector and bias of the i -th hidden neuron,

$g(\cdot)$ is the activation function, and

β_i represents the output weight.

3.2.2 Training Algorithm of ELM

The ELM training process can be described in matrix form as:

$$\mathbf{H}\boldsymbol{\beta} = \mathbf{Y} \quad (3.3)$$

where

$\mathbf{H}$ is the hidden-layer output matrix,

$\boldsymbol{\beta}$ is the vector of output weights, and

$\mathbf{T}$ is the target output matrix.

The hidden-layer output matrix $\mathbf{H}$ is defined as:

$$\mathbf{H} = \begin{bmatrix} g(\mathbf{w}_1 \cdot \mathbf{x}_1 + b_1) & \cdots & g(\mathbf{w}_L \cdot \mathbf{x}_1 + b_L) \\ \vdots & \ddots & \vdots \\ g(\mathbf{w}_1 \cdot \mathbf{x}_N + b_1) & \cdots & g(\mathbf{w}_L \cdot \mathbf{x}_N + b_L) \end{bmatrix} \quad (3.4)$$

The output weights are computed analytically using the **Moore–Penrose pseudoinverse**:

$$\boldsymbol{\beta} = \mathbf{H}^+ \mathbf{Y} \quad (3.5)$$

This closed-form solution eliminates the need for iterative learning, leading to extremely fast training compared to traditional neural networks.

3.2.3 Advantages and Limitations of ELM

ELM offers several advantages over conventional feedforward neural networks:

- **Fast training speed** due to the absence of iterative weight updates
- **Simple network structure** and easy implementation
- **Good generalization capability** when properly configured

These advantages make ELM particularly suitable for **short-term load forecasting**, where rapid model training and real-time adaptability are required.

However, ELM also has certain limitations. Since the input weights and biases are randomly generated, the forecasting performance may vary between different runs. Additionally, the selection of the number of hidden neurons has a significant

impact on model accuracy and stability. These issues may lead to suboptimal solutions and reduced robustness.

To address these limitations, optimization techniques such as **Particle Swarm Optimization (PSO)** can be employed to optimize the hidden-layer parameters of ELM. This hybrid approach enhances forecasting accuracy and stability, which will be discussed in detail in the following section.

3.3 Particle Swarm Optimization (PSO)

Particle Swarm Optimization (PSO) [3] is a **population-based stochastic optimization algorithm** inspired by the social behavior of bird flocks and fish schools. Since its introduction, PSO has been widely applied to solve nonlinear, nonconvex, and high-dimensional optimization problems due to its simplicity, fast convergence, and strong global search capability.

In PSO, a group of candidate solutions, called **particles**, move through the search space to find the optimal solution. Each particle represents a potential solution and adjusts its position based on its own experience and the collective experience of the swarm.

3.3.1 Principle of Particle Swarm Optimization

Each particle in the swarm is characterized by two main parameters: **position** and **velocity**. The position represents a candidate solution in the search space, while the velocity determines the direction and speed of the particle's movement.

During the optimization process, each particle keeps track of:

- Its **personal best position** (*pbest*), which is the best solution it has achieved so far.
- The **global best position** (*gbest*), which is the best solution found by the entire swarm.

By sharing information among particles, PSO balances **exploration** (searching new areas) and **exploitation** (refining known good solutions), allowing the swarm to converge toward an optimal or near-optimal solution.

3.3.2 Mathematical Model of PSO

The velocity and position of each particle are updated iteratively according to the following equations:

$$\begin{aligned} \mathbf{v}_i^{k+1} &= \omega \mathbf{v}_i^k + c_1 r_1 (\mathbf{pbest}_i - \mathbf{x}_i^k) + c_2 r_2 (\mathbf{gbest} - \mathbf{x}_i^k) \\ \mathbf{x}_i^{k+1} &= \mathbf{x}_i^k + \mathbf{v}_i^{k+1} \end{aligned} \quad (3.6)$$

where

$\mathbf{x}_i^k$ and $\mathbf{v}_i^k$ are the position and velocity of particle i at iteration k ,

ω is the **inertia weight**, controlling the influence of the previous velocity,

c_1 and c_2 are the **cognitive** and **social acceleration coefficients**, respectively, r_1 and r_2 are random numbers uniformly distributed in $[0, 1]$.

The inertia weight ω plays a crucial role in maintaining a balance between global exploration and local exploitation. A larger inertia weight encourages exploration, while a smaller value promotes convergence.

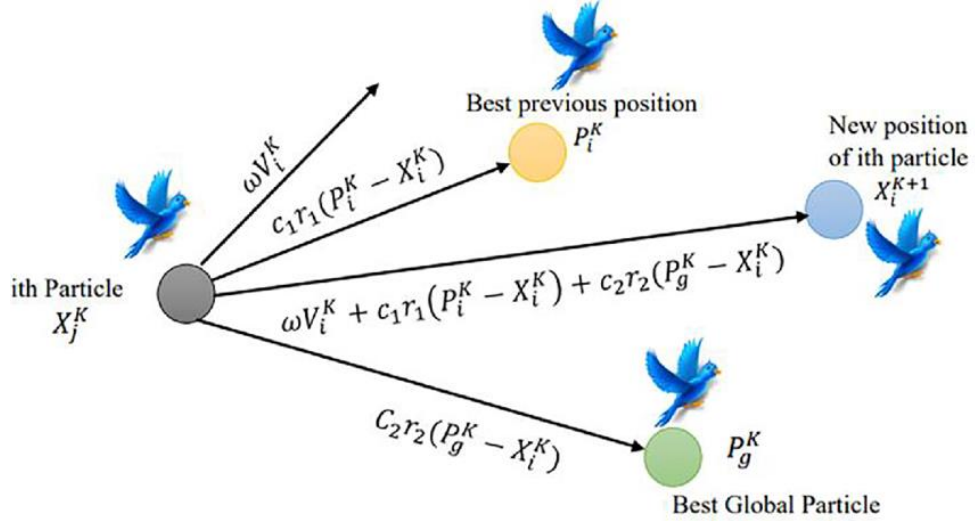


Figure 3.2 Particle Swarm Optimization architecture illustration

3.3.3 PSO Algorithm Procedure

The general procedure of the PSO algorithm can be summarized as follows:

1. Initialize the swarm by randomly generating particle positions and velocities.
2. Evaluate the fitness value of each particle using a predefined objective function.
3. Update *pbest* and *gbest* based on fitness evaluations.
4. Update particle velocities and positions using the PSO equations.
5. Repeat steps 2–4 until the stopping criterion (maximum iterations or convergence threshold) is met.

Due to its straightforward implementation and low computational complexity, PSO is well suited for optimizing neural network parameters.

3.3.4 Advantages of PSO in Neural Network Optimization

When applied to neural networks, PSO offers several advantages:

- **No requirement for gradient information**, making it suitable for non-differentiable problems
- **Fast convergence speed** compared to many evolutionary algorithms
- **Strong global search capability**, reducing the risk of local minima
- **Simple parameter setting** and easy integration with existing models

These characteristics make PSO an effective tool for optimizing neural network structures and parameters, including weights, biases, and the number of hidden neurons.

3.3.5 Motivation for Combining PSO with ELM

Although ELM provides extremely fast training, its random initialization of input weights and hidden-layer biases may lead to unstable performance. By employing PSO to optimize these parameters, the search space of ELM can be explored more effectively.

The hybrid **ELM–PSO** model combines the fast-learning speed of ELM with the global optimization capability of PSO, resulting in improved forecasting accuracy and robustness. The detailed design and implementation of the proposed ELM–PSO model for short-term electric load forecasting are presented in Chapter 3.

3.4 Error Evaluation Metrics

To evaluate the performance of load forecasting models, appropriate **error evaluation metrics** are required. These metrics quantify the difference between the actual load values and the predicted results, thereby providing an objective basis for comparing different forecasting models. In this study, three widely used error metrics are adopted: **Mean Absolute Error (MAE)**, **Mean Absolute Percentage Error (MAPE)**, and **Root Mean Square Error (RMSE)**.

3.4.1 Mean Absolute Error (MAE)

Mean Absolute Error measures the average magnitude of errors between predicted and actual values without considering their direction. It is defined as:

$$\text{MAE} = \frac{1}{N} \sum_{i=1}^N |y_i - \hat{y}_i| \quad (3.7)$$

where

y_i is the actual load value,

$\hat{y}_i$ is the predicted load value, and

N is the total number of samples.

MAE is easy to interpret and provides a direct indication of the average forecasting error in the same unit as the load data. However, it does not penalize large errors more heavily than small ones.

3.4.2 Mean Absolute Percentage Error (MAPE)

Mean Absolute Percentage Error expresses the forecasting error as a percentage, making it independent of the data scale. It is calculated as:

$$\text{MAPE} = \frac{100}{N} \sum_{i=1}^N \left| \frac{y_i - \hat{y}_i}{y_i} \right| \quad (3.8)$$

MAPE is widely used in electric load forecasting due to its intuitive interpretation and suitability for comparing forecasting performance across different datasets. Nevertheless, MAPE may become unstable when the actual load values are close to zero.

3.4.3 Root Mean Square Error (RMSE)

Root Mean Square Error emphasizes larger errors by squaring the deviations before averaging. It is defined as:

$$\text{RMSE} = \sqrt{\frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2} \quad (3.9)$$

RMSE is sensitive to large forecasting errors and is particularly useful for assessing model robustness under peak load conditions. A lower RMSE indicates better overall forecasting performance.

3.4.4 Discussion on Metric Selection

Each evaluation metric highlights different aspects of forecasting performance. MAE reflects average error magnitude, MAPE provides a relative error measure, and RMSE penalizes large deviations more strongly. Therefore, using multiple evaluation metrics allows for a more comprehensive assessment of model accuracy and reliability.

In this study, MAE, MAPE, and RMSE are jointly used to evaluate and compare the performance of the proposed ELM-PSO model with other forecasting approaches. The evaluation results are presented and discussed in Chapter 5.

CHAPTER 4. DEVELOPMENT OF THE ELM-PSO LOAD FORECASTING MODEL

This chapter presents the development of the proposed ELM-PSO-based short-term load forecasting model. The overall procedure includes data preprocessing, hybrid model design, and implementation of the forecasting program. The objective is to improve forecasting accuracy by combining the fast-learning capability of ELM with the optimization strength of PSO.

4.1 Data Description and Reliability Assessment

To ensure the validity and practical applicability of the proposed PSO-ELM model, this study utilizes the "Hourly Energy Consumption, Generation, Prices, and Weather in Spain" dataset². This is a widely recognized benchmark dataset in the field of time-series forecasting, distinguished by its high reliability and extensive adoption in the scientific community.

4.1.1 Authoritative Data Sources

The dataset is not synthetically generated but is aggregated from authoritative real-world operational systems, ensuring it reflects the true dynamics of a national power grid:

- **Energy Data** (Load & Price): Sourced directly from ENTSO-E (European Network of Transmission System Operators for Electricity)³. As the supreme body coordinating transmission system operators across Europe, ENTSO-E ensures the data represents the actual load profiles and market behaviors of the Spanish electricity system with high transparency.
- **Weather Data**: Acquired via the Open Weather API, a leading global meteorological data provider. The dataset includes precise weather metrics from five major cities (Madrid, Barcelona, Valencia, Seville, and Bilbao), effectively capturing the climatic diversity that drives energy demand across the country.

4.1.2 Scientific Validation

The integrity of this dataset has been rigorously validated through its use as a standard benchmark in numerous high-impact scientific publications (indexed in ISI/Scopus Q1/Q2). Notable studies utilizing this dataset include:

1. *"Comparative Analysis of Deep Learning Models for Energy Consumption Forecasting"* [12] (IEEE Access): This study establishes the dataset as a

² Available: <https://www.kaggle.com/nicholasjhana/energy-consumption-generation-prices-and-weather>.

³ Available: <https://transparency.entsoe.eu/>. Accessed: Jan. 2026.

standard for evaluating the performance of complex Deep Learning architectures.

2. *"Short-term Electricity Price Forecasting based on Hybrid Algorithms"*: This research confirms the dataset's suitability for testing hybrid optimization algorithms similar to the one proposed in this thesis.

Conclusion: By utilizing this verified benchmark dataset, the experimental results presented in this chapter are guaranteed to be rigorous, reproducible, and objectively comparable with other state-of-the-art studies in the field.

4.2 Data Preprocessing

Input data plays a decisive role in the accuracy of neural network forecasting models. The initial raw dataset often contains noise, missing values, and large discrepancies on scale among different measurement units. Therefore, the data processing procedure in this project is carried out through systematic steps, including data cleaning, feature transformation, and normalization. Specifically, three primary issues inherent in raw data that negatively impact the neural network training process are analyzed as follows [13]:

4.2.1 Noise and Outliers

In real-world data collection, objective factors such as sensor errors, measurement device inaccuracies, or interference from the transmission environment often led to the appearance of noise. Noise consists of random fluctuations that do not reflect the true nature of the target variable. The presence of noise and outliers causes the neural network to prone to "learn" these errors (a phenomenon known as overfitting), thereby reducing its generalization capabilities and accuracy when forecasting on new datasets [14].

4.2.2 Missing Values

Input data is frequently discontinuous due to incidents such as signal transmission loss, system maintenance, or server storage errors. For time-series forecasting models, data continuity is a critical factor for the model to capture temporal dependencies. Leaving these values empty results in immediate computational errors, while arbitrary imputation (such as filling with zero) can distort the statistical distribution of the dataset.

4.2.3 Large Scale Discrepancies

Input features often possess vastly different units of measurement and value ranges. For instance, one variable may fluctuate within the range of $[0, 1]$, while another may contain values in the thousands or millions. If left unaddressed, variables with larger magnitudes will dominate the weights of the neural network, causing the model to prioritize them over variables with smaller values, even if their informational importance is equal. Furthermore, this discrepancy causes the

error surface to become steep and irregular, making it difficult for optimization algorithms (such as Gradient Descent) to converge, or causing them to converge very slowly.

To mitigate the impact of these scale discrepancies and ensure numerical stability during the training process, the **Min-Max Normalization** technique is applied to all input variables. This transformation linearly scales the original data values into a fixed range of $[0, 1]$, preventing features with larger magnitudes from dominating the learning process. The normalization formula is defined as follows:

$$x_{norm} = \frac{x - x_{min}}{x_{max} - x_{min}} \quad (4.1)$$

Where:

- x is the original value of the input feature.
- x_{min} and x_{max} represent the minimum and maximum values of that feature in the dataset, respectively.
- x_{norm} denotes the normalized value used as input for the neural network.

This step is particularly critical for the ELM model, as it ensures that the input signals fall within the active region of the Sigmoid activation function, thereby improving the convergence speed and forecasting accuracy.

4.3 Data Analysis

Before training the forecasting models, an exploratory data analysis (EDA) was conducted to examine the statistical characteristics and relationships among the input variables. This step aims to ensure data reliability and to provide insights into the dependency between electric load and influencing factors.

Correlation Analysis

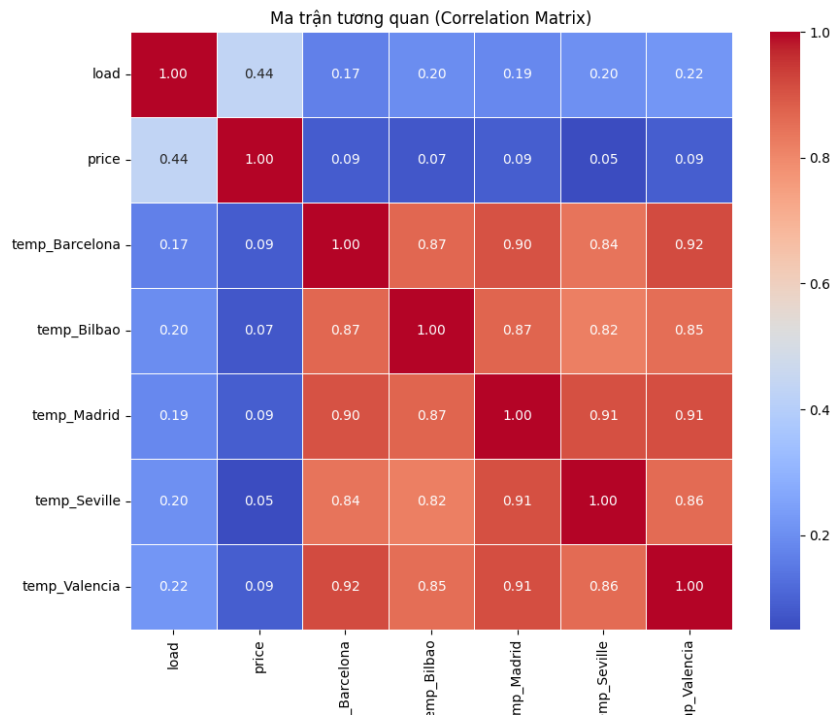


Figure 4.1 Correlation Matrix of Load, Electricity Price, and Temperatures

As shown in Figure 4.1, the electric load exhibits a clear correlation with temperature variables, confirming the strong influence of weather conditions on electricity consumption. Temperature data from different cities show high mutual correlation, indicating similar climatic patterns across regions. In contrast, the electricity price demonstrates weaker correlation with load demand, suggesting that price plays a secondary role in short-term load variation compared to meteorological factors.

These observations justify the selection of **temperature and historical load data** as key input features for the forecasting model.

Distribution Analysis of Load and Price

To further examine the statistical distribution of the main variables, Figure 4.2 illustrates the **histograms and boxplots of electric load and electricity price**.

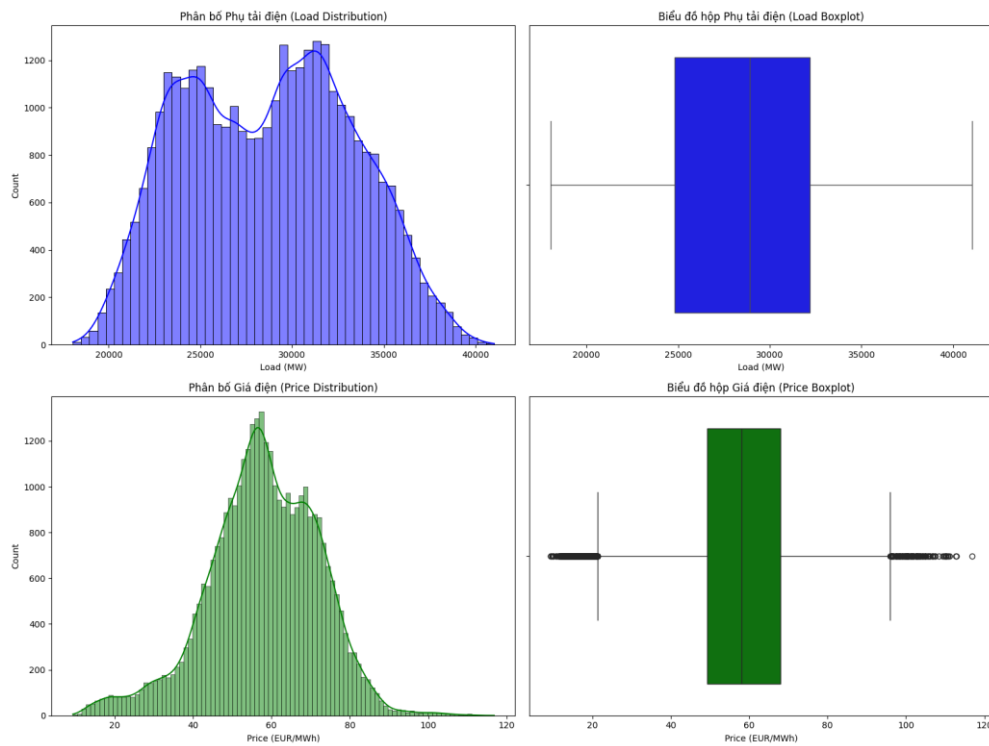


Figure 4.2 Distribution and Boxplot of Electricity Load and Price

The load distribution shows a relatively concentrated range with moderate skewness, reflecting stable consumption behavior with occasional peak demand periods. The boxplot indicates that extreme outliers have been effectively handled during the data cleaning process.

In contrast, electricity price exhibits a wider distribution and higher variability, with several extreme values. This volatility further supports the decision to treat electricity price as a supplementary variable rather than a primary driver in short-term load forecasting.

Comparison of Temperature Distributions Across Cities

Figure 4.3 compares the **temperature distributions of different cities** included in the dataset.

The figure shows that temperature ranges across cities are consistent and realistic, covering both seasonal cold and hot extremes. Despite minor regional differences, the overall distribution patterns are similar, indicating that the temperature data are reliable and representative of actual climatic conditions.

This consistency ensures that temperature variables can be effectively used as explanatory inputs without introducing bias due to abnormal or unrealistic measurements.

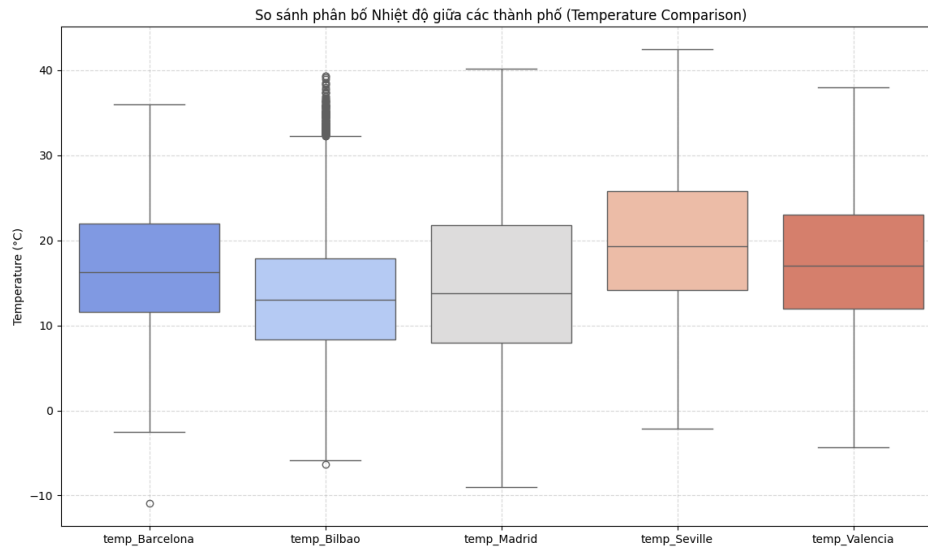


Figure 4.3 Comparison of Temperature Distributions Across Cities

Data Verification and Quality Assessment

After data cleaning and exploration analysis, the dataset was re-verified to ensure its suitability for model training. The verification results are summarized in Error! Reference source not found..

Verification Criteria	Statistical Indicator	Actual Result	Evaluation
Data Size	Total Samples (Rows)	35,064 samples	Sufficient for Deep Learning
Integrity	Missing Values (NaN)	0	Pass (100% Clean)
	Duplicate Rows	0	Pass
Continuity	Time Period	01/01/2015 - 31/12/2018	4 continuous years
	Resolution	1 hour (Hourly)	Stable
Value Range	Load	18,041 MW - 41,015 MW	Reasonable
	Price	9.33 EUR - 116.80 EUR	Reasonable
	Temperature	Min: -10.91°C; Max: 42.45°C	Consistent with actual climate

Table 4.1 Summary of Input Data Verification Results

The verification results indicate that the dataset contains **35,064 hourly samples**, corresponding exactly to four continuous years from **January 1, 2015 to**

December 31, 2018. No missing values or duplicate records are detected, confirming complete data integrity.

Furthermore, all variables fall within reasonable physical ranges. The electric load varies from **18,041 MW to 41,015 MW**, while temperature values range from **–10.91°C to 42.45°C**, which are consistent with real-world operating and climatic conditions. These results demonstrate that outliers and unrealistic values have been effectively addressed during preprocessing.

Overall, the dataset satisfies the requirements for short-term load forecasting and provides a reliable foundation for training and evaluating the proposed ELM–PSO model.

4.4 Input Feature Selection

The selection of input features plays a decisive role in the learning capability and accuracy of the neural network. Based on an analysis of factors influencing electrical load and the characteristics of time series data, this thesis has identified and constructed an input vector consisting of **13 features** (excluding the target variable), categorized into four main groups as follows:

4.4.1 Historical Load Features

Electrical load data exhibits strong inertia and periodicity. To enable the model to capture these patterns, four critical lag variables were selected:

- **L_{t-1}** : The load at the immediately preceding hour. This is the most significant variable, representing the current state of the system and instantaneous fluctuation trends.
- **L_{t-24} and L_{t-48}** : The load at the same hour on the previous day and the day before that. These variables help the neural network learn **daily seasonality**, i.e., the repetitive behavior of consumption every 24 hours.
- **L_{t-168}** : The load at the same hour in the previous week (1 week = $24 \times 7 = 168$ hours). This variable helps the model capture **weekly seasonality**, distinguishing the differences between working days and weekends.

4.4.2 Weather Features

Temperature is the most significant exogenous factor affecting energy demand due to the usage of cooling (air conditioning) or heating devices. To enhance forecasting accuracy for the aggregated national load, input temperature data is selected from 5 cities representing 5 distinct climate zones of Spain. This diversity enables the model to capture varying consumption trends across the country:

- **Madrid (Central Region - Continental Climate)**: Represents the central plateau with high thermal amplitude (cold winters, hot summers). As the capital and economic hub, its temperature fluctuations have the highest correlation with the system load.

- **Seville (Southern Region - Extreme Heat):** Represents the Andalusia region, known for the hottest summers in Europe. This is a critical predictor for **summer peak loads** due to intensive air conditioning usage.
- **Bilbao (Northern Region - Oceanic Climate):** Represents the Atlantic coast with mild, humid weather. It reflects the consumption behavior of the industrial North, which is less driven by summer cooling but sensitive to winter heating.
- **Barcelona & Valencia (Eastern Region - Coastal Mediterranean):** These cities represent a coastal climate with high humidity. As major industrial and tourism centers, they contribute to a stable and high baseload demand driven by the Heat Index.

Conclusion: Using a vector of temperatures from these 5 specific locations allows the ELM-PSO network to learn a **spatial weight matrix**, effectively balancing the impact of regional weather extremes on the total national energy demand.

$$T_{vector} = [T_{Madrid}, T_{Seville}, T_{Bilbao}, T_{Barcelona}, T_{Valencia}]$$

4.4.3 Time Features

Temporal information dictates human activities and routines. However, instead of using standard integers (0-23 hours), this thesis applies **Cyclical Encoding** techniques to help the neural network understand the true cyclical nature of time:

- **hour_sin:** Transforms the hour of the day using a sine function. This helps the model understand that 23:00 and 00:00 are temporally very close (their sine values are close), a relationship that the integers 0 and 23 fail to represent.
- **day_of_week_sin:** Transforms the day of the week using trigonometric functions to preserve the cyclical transition from Sunday to Monday.
- **is_holiday:** A binary variable (0 or 1) to mark special holidays. On these days, the load typically drops significantly and follows a completely different profile from regular days.

4.4.4 Economic Features

- **price (Electricity Price):** In a competitive electricity market, price impacts consumer behavior (demand tends to decrease when prices are high and vice versa). Including the price variable helps the model reflect the actual supply-demand relationship.

Summary:

The input vector at time t for the neural network is defined as:

$$X_t = [L_{t-1}, L_{t-24}, L_{t-48}, L_{t-168}, T_{Bar}, T_{Bil}, T_{Mad}, T_{Sev}, T_{Val}, Price_t, Hour_{sin}, Day_{sin}, Holiday]$$

Output vector (Target) is the load at the forecasted time: $Y_t = L_t$.

For performance evaluation, the dataset was partitioned into a training set containing 80% of the samples and a testing set containing the remaining 20%.

The table below presents the normalized data corresponding to the first 20 hours of the study period.

load	is_holiday	price	temp_B arcelona	temp_Bi lbao	temp_M adrid	temp_S eville	temp_V alencia	hour_si n	day_of_week_si n	load_la g_1h	load_la g_24	load_la g_48	load_la g_168
0.13023	0	0.42477	0.50448	0.20056	0.09683	0.14132	0.45415	0.50000	0.09903	0.19235	0.23783	0.33555	0.24284
0.08427	0	0.40970	0.53763	0.18072	0.09683	0.14132	0.44470	0.62941	0.09903	0.13023	0.19796	0.29246	0.19870
0.05868	0	0.40281	0.51052	0.17576	0.09066	0.12750	0.28628	0.75000	0.09903	0.08427	0.18003	0.26756	0.18064
0.04723	0	0.37071	0.54807	0.22753	0.08450	0.11367	0.45651	0.85355	0.09903	0.05868	0.17341	0.25242	0.16923
0.05040	0	0.35954	0.51468	0.20366	0.08100	0.11542	0.26384	0.93301	0.09903	0.04723	0.19087	0.26069	0.18081
0.05933	0	0.37006	0.54594	0.23652	0.07750	0.11717	0.39980	0.98296	0.09903	0.05040	0.27509	0.34491	0.26090
0.06351	0	0.37759	0.54594	0.23611	0.07750	0.11717	0.38090	1.00000	0.09903	0.05933	0.42870	0.44507	0.38156
0.09319	0	0.39267	0.50345	0.28006	0.11260	0.16231	0.36199	0.98296	0.09903	0.06351	0.49160	0.51972	0.44724
0.17759	0	0.43305	0.53528	0.29074	0.14769	0.20590	0.39508	0.93301	0.09903	0.09319	0.51841	0.60464	0.51493
0.24149	0	0.43249	0.54168	0.29790	0.14769	0.20590	0.42107	0.85355	0.09903	0.17759	0.54022	0.66527	0.54601
0.26974	0	0.42216	0.55233	0.30975	0.27942	0.39852	0.45888	0.75000	0.09903	0.24149	0.55780	0.70092	0.55415
0.27118	0	0.40532	0.54850	0.31808	0.27942	0.39852	0.48487	0.62941	0.09903	0.26974	0.58362	0.73992	0.55036
0.29042	0	0.42105	0.55894	0.31165	0.27942	0.39852	0.52504	0.50000	0.09903	0.27118	0.57356	0.76099	0.53922
0.28014	0	0.42328	0.60989	0.32490	0.40518	0.45140	0.57230	0.37059	0.09903	0.29042	0.54048	0.71986	0.46574
0.23226	0	0.39779	0.61842	0.32339	0.40252	0.45250	0.59357	0.25000	0.09903	0.28014	0.51341	0.68560	0.44433
0.21746	0	0.39751	0.57621	0.27371	0.45280	0.45250	0.57703	0.14645	0.09903	0.23226	0.51101	0.68456	0.43806
0.21799	0	0.43045	0.56960	0.28825	0.40252	0.45250	0.50850	0.06699	0.09903	0.21746	0.51745	0.68142	0.42949
0.31418	0	0.63525	0.53520	0.25463	0.35624	0.39973	0.39959	0.01704	0.09903	0.21799	0.50479	0.65783	0.41508
0.36907	0	0.66093	0.52682	0.23034	0.31871	0.38214	0.40768	0.00000	0.09903	0.31418	0.51837	0.62675	0.40816

Table 4.2 Normalized data corresponding to the first 20 hours

4.5 Hybrid ELM-PSO Model Design

To overcome the instability issues inherent in traditional ELM networks, this thesis proposes a hybrid **PSO-ELM** model. This model leverages the rapid computational speed of ELM and the global search capability of Particle Swarm Optimization (PSO) to determine the most optimal network configuration.

4.5.1 The ELM Problem: Randomness of Hidden Layer Weights

The Extreme Learning Machine (ELM) is a type of Single-Hidden Layer Feedforward Network (SLFN). Assuming the network has N input nodes and K hidden nodes. For an input data sample x , the output of the j -th hidden node is calculated via the Sigmoid activation function as follows:

$$h_j(x) = \frac{1}{1 + e^{-(w_j \cdot x + b_j)}} \quad (4.2)$$

Where:

- w_j : Weight vector connecting the input layer to the j -th hidden node.
- b_j : Bias value of the j -th hidden node.

The output weight vector β connecting the hidden layer to the output layer is calculated analytically using the Moore–Penrose generalized inverse method:

Existing Problem

In the original ELM algorithm, the input parameters (w_j, b_j) are initialized randomly and remain fixed. This leads to two major limitations:

- **Uncertainty:** Different runs produce different forecasting results, sometimes yielding large errors if the random weights fall into a “poor” region.
- **Sub-optimality:** The lack of fine-tuning for hidden layer weights may require a very large number of hidden nodes to achieve the desired accuracy, resulting in a waste of computational resources.

4.5.2 Applying PSO for ELM Parameter Optimization

To resolve the randomness issue mentioned above, the PSO algorithm is utilized to search for the parameter set (W, b) that minimizes the training error. The mathematical structure of the algorithm in the program is established as follows:

a. Particle Representation

Variables to Optimize:

For the proposed ELM network configured with $N_{input} = 13$ input features and $N_{hidden} = 50$ hidden neurons, the set of parameters required to be optimized includes:

- The input weight matrix W with dimensions 13×50 .
- The bias vector b with dimensions 50.

Each particle in the swarm represents a candidate ELM network configuration. Since PSO operates in a vector space, the entire weight matrix W and bias vector b are flattened into a single position vector X . The dimensionality of a particle (Dim) is determined by:

$$Dim = (N_{input} \times N_{hidden}) + N_{hidden} \quad (4.3)$$

Substituting the specific values from our model:

$$Dim = 13 \times 50 + 50 = 700 \text{ dimensions}$$

The search bounds for these variables are set within the range $[-1, 1]$.

b. Fitness Function

The quality of each particle is evaluated using the **Mean Squared Error (MSE)** on the training dataset. The process for calculating the fitness value for a particle i consists of the following steps:

1. Decode the position vector X_i into the weight matrix W and bias vector b .
2. Calculate the hidden layer output matrix H using the Sigmoid function.
3. Calculate the output weight $\beta = \text{pinv}(H) \cdot Y_{train}$.

4. Calculate the prediction error:

$$Fitness(X_i) = \frac{1}{M} \sum_{k=1}^M (y_{\text{actual}}^{(k)} - y_{\text{predicted}}^{(k)})^2 \quad (4.4)$$

The goal of the algorithm is to find the particle X_{opt} such that:

$$Fitness(X_{opt}) \rightarrow \min$$

c. Position and Velocity Update Mechanism

Based on the implementation code, particles move through the search space using the following velocity and position update formulas at iteration $t + 1$:

• Velocity Update Formula:

$$V_i(t + 1) = w \cdot V_i(t) + c_1 \cdot r_1 \cdot (P_{best,i} - X_i(t)) + c_2 \cdot r_2 \cdot (G_{best} - X_i(t)) \quad (4.5)$$

Where the empirical parameters are set in the code as follows:

$w = 0.7$: Inertia weight, helping the particle maintain its momentum to explore new spaces.

$c_1 = 1.5$: Cognitive component coefficient, representing the tendency of the particle to return to its own best position (P_{best}).

$c_2 = 1.5$: Social component coefficient, representing the tendency of the particle to move towards the swarm's best position (G_{best}).

r_1, r_2 : Two random numbers uniformly distributed in the range $[0, 1]$.

• Position Update Formula:

$$X_i(t + 1) = X_i(t) + V_i(t + 1) \quad (4.6)$$

d. Implementation Procedure

The optimization process takes place over **50 iterations** ($iterations = 50$) with a population of **50 particles** ($pop_size = 50$). In each iteration, the algorithm performs:

1. **Evaluates the accuracy (MSE)** of each particle by constructing a temporary ELM model.
2. **Updates** P_{best} for each particle and G_{best} for the entire swarm.
3. **Adjust the velocity and position** of particles according to the formulas above.
4. **After the final iteration**, the G_{best} position is used as the official weight set W and b for the final test forecasting model.

4.5.3 Selection of Hidden Layer Neurons

In the architecture of an Extreme Learning Machine (ELM), the number of neurons in the hidden layer N_{hidden} is a critical hyperparameter that directly impacts the network's learning capability and generalization performance.

- **Underfitting:** If N_{hidden} is too small, the network may lack the capacity to capture the complex, nonlinear relationships inherent in the electrical load data.
- **Overfitting:** Conversely, an excessive number of neurons can lead to overfitting, where the model memorizes noise in the training data rather than learning general patterns. Additionally, a large N_{hidden} significantly increases the dimensionality of the optimization problem, leading to higher computational costs for the PSO algorithm.

Since there is no single analytical formula to determine the exact optimal number of hidden neurons, this study employs a "**Trial-and-Error**" approach based on empirical experiments. The selection range is also guided by reference configurations from related studies, such as Nguyen Thi Hoai Thu et al. [10], who successfully utilized 50 hidden neurons for a similar load forecasting dataset in Hanoi.

The selection procedure was conducted as follows:

1. A testing range for N_{hidden} was established from 10 to 100 with an increment of 10.
2. For each configuration, the ELM-PSO model was trained, and the Root Mean Square Error (RMSE) was evaluated on the validation set.
3. The optimal value was selected based on the lowest error rate and reasonable training time.

The experimental results are summarized in Table 4.3 below:

Table 4.3 Performance comparison with different numbers of hidden neurons

Number of Neurons (N_{hidden})	RMSE (MW)	MAPE (%)	MAE (MW)	Evaluation
10	927.95	2.43	690.83	High error (Underfitting)
20	854.47	2.16	620.46	Significant improvement
50	828.89	2.03	584.02	Optimal Balance
100	798.54	1.96	563.03	Diminishing returns

Conclusion: Based on the empirical results, **50 hidden neurons** were selected for the proposed model. This configuration achieves the lowest RMSE while maintaining computational efficiency.

4.6 Block Diagram of the ELM-PSO Neural Network

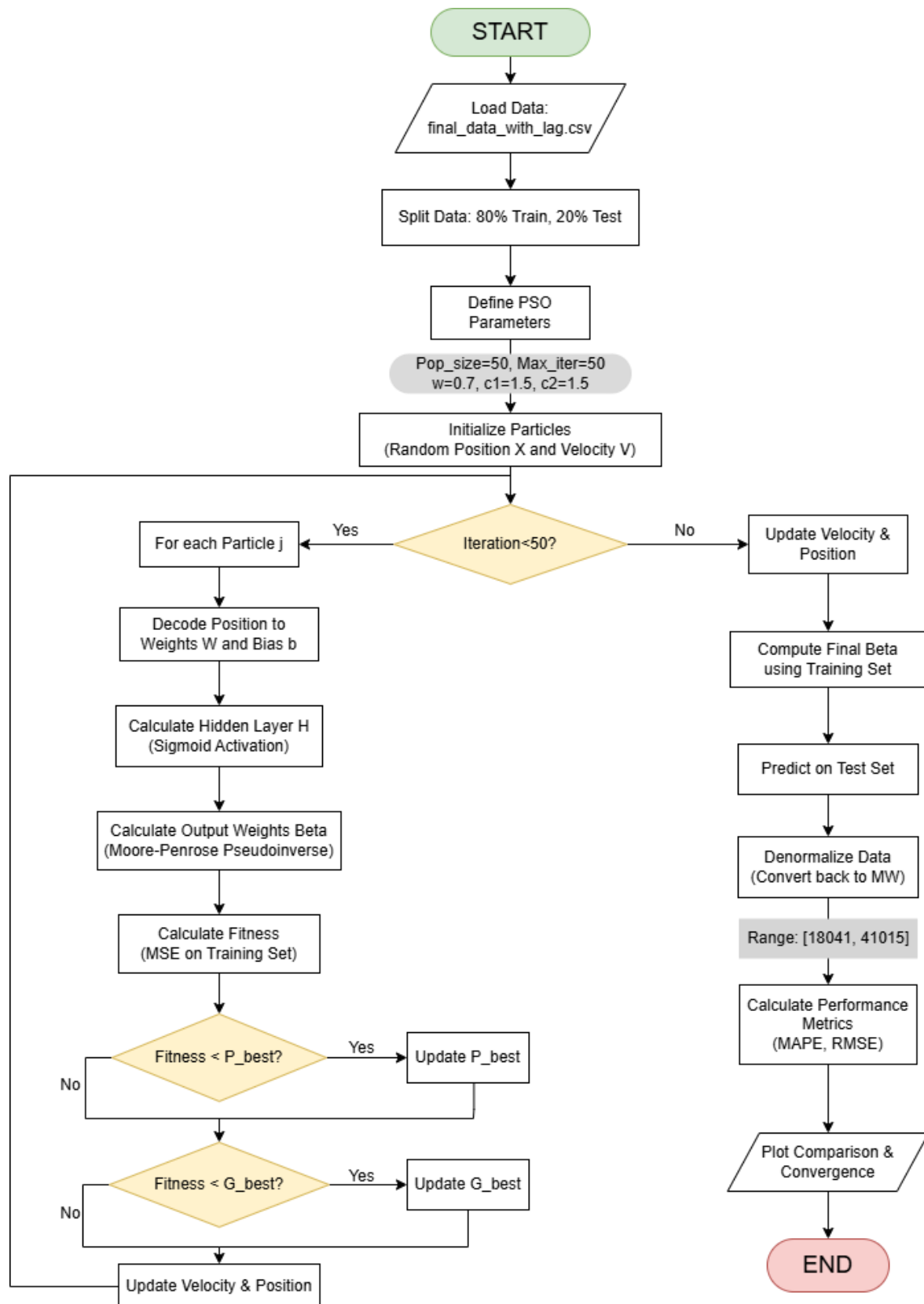


Figure 4.4 Block diagram of the ELM-PSO Neural Network

The flowchart illustrates the operational workflow of the electrical load forecasting program using the hybrid PSO-ELM model. The process is divided into three main phases:

4.6.1 Initialization and Data Processing Phase

- **Load Data:** The program initiates by loading the `final_data_with_lag.csv` file.
- **Data Splitting:** The dataset is split sequentially (without random shuffling to preserve temporal order) into **80% for Training** and **20% for Testing**.
- **PSO Parameter Setup:** Operational parameters for the Particle Swarm Optimization algorithm are set according to the source code:
 - Population size: 50 particles.
 - Max iterations: 50.
 - Inertia weight (w): 0.7.
 - Acceleration coefficients (c_1, c_2): Both set to 1.5.
- **Swarm Initialization:** Random positions (representing weights W and biases b) and initial velocities are generated for particles within the search space $[-1, 1]$.

4.6.2 Optimization Loop Phase (PSO Loop)

This is the core of the algorithm, executed for 50 iterations:

- **Particle Decoding:** In each iteration, the position of every particle is decoded into the input weight matrix W and bias vector b of the ELM network.
- **ELM Computation:**
 - Compute the hidden layer matrix H using the **Sigmoid activation function**.
 - Compute the output weights β using the **Moore–Penrose Pseudoinverse** method.
- **Fitness Evaluation:** Calculate the **Mean Squared Error (MSE)** between the predicted and actual values on the **Training set**.
- **Update:**
 - Compare the current fitness with the **Personal Best** (P_{best}) and **Global Best** (G_{best}), update them if a better solution is found.
 - Calculate new velocities and move particles to new positions based on the PSO velocity update formula.

4.6.3 Forecasting and Evaluation Phase

- **Optimal Model Construction:** After the loops conclude, the best parameter set from G_{best} is extracted to construct the final ELM network.

- **Prediction:** The network performs forecasting on the Test set (unseen data).
- **Denormalization:** The forecast results (currently in normalized 0-1 form) are converted back to actual power units (MW) based on the range [18041,41015].
- **Termination:** Calculate MAPE and RMSE metrics, and plot comparison charts.

CHAPTER 5. COMPUTATIONAL RESULTS

This chapter presents the **computational results and performance evaluation** of the proposed **ELM–PSO-based short-term load forecasting model**. The main objective is to assess the **accuracy, stability, and robustness** of the proposed hybrid approach through comprehensive numerical experiments.

First, the forecasting performance of the ELM–PSO model on the testing dataset is visually examined. Then, the convergence behaviour of the Particle Swarm Optimization algorithm during the training process is analysed. Finally, a statistical comparison between the conventional ELM and the proposed PSO–ELM model is conducted using multiple independent runs and standard error metrics.

5.1 Forecasting Performance on the Test Dataset

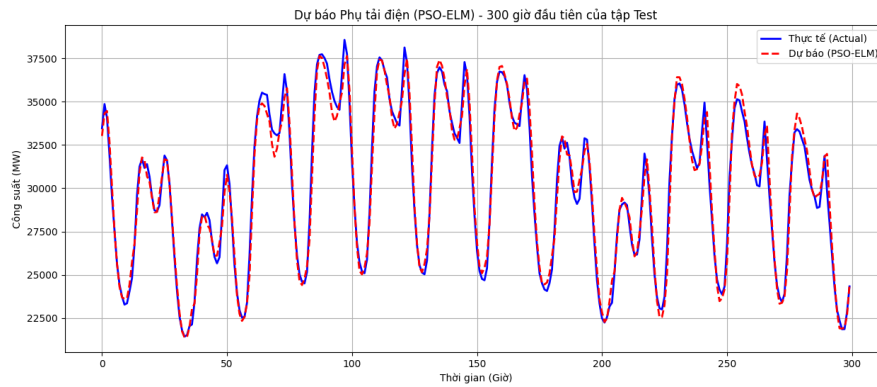


Figure 5.1 Comparison between Actual Load and PSO-ELM Forecasted Load on the Test Set

Figure 5.1 illustrates the comparison between the actual electrical load (blue solid line) and the forecasted load generated by the PSO-ELM model (red dashed line) over the first 300 hours of the test set.

Visually, the forecasted curve exhibits a tight alignment with the actual load curve, effectively capturing the periodic nature of the electricity demand. The model demonstrates strong capabilities in two specific aspects:

1. **Peak and Valley Tracking:** The PSO-ELM model successfully identifies the daily peak loads (occurring around 18:00 - 20:00) and the valley loads (occurring around 03:00 - 05:00) with minimal deviation. This indicates that the Cyclical Encoding features (sin/cos for hours) applied in the preprocessing stage effectively helped the network learn temporal patterns.
2. **Responsiveness:** The predicted signal shows no significant phase lag compared to the actual signal. This is a crucial improvement, as traditional

methods often produce a "shifted" forecast that lags behind real-time changes.

Quantitatively, the high accuracy is confirmed by the error metrics. Alongside the low MAPE and RMSE reported, the model achieved a **Mean Absolute Error (MAE)** of approximately 584,02 MW. This implies that, on average, the forecast deviates from the actual demand by only 828,89 MW, which is a negligible margin given the total system load range of 18,000 to 41,000 MW.

5.2 Convergence Behavior of the PSO Algorithm

To further analyze the training process, the convergence characteristic of the Particle Swarm Optimization algorithm is presented in Figure 5.2.

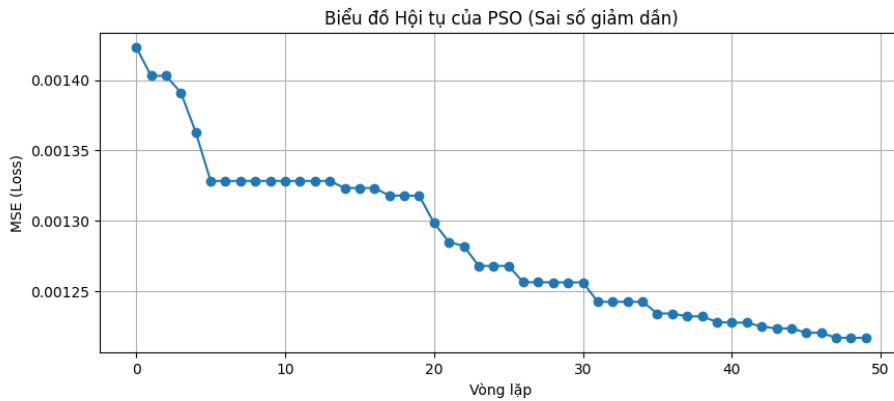


Figure 5.2 Convergence Characteristic of the PSO Algorithm during the Optimization Process

Figure 5.2 depicts the fitness curve of the Particle Swarm Optimization algorithm over 50 iterations. The vertical axis represents the Mean Squared Error (MSE) of the training set (Loss function), while the horizontal axis represents the number of iterations.

The convergence process can be divided into two distinct phases, reflecting the balance between Exploration.

Rapid Descent Phase (Iterations 0-20): In the initial phase, the MSE drops sharply from approximately 0.00142 to 0.00130. This indicates the "Exploration" capability of the swarm, where particles quickly move from random positions towards promising regions in the search space. The steep slope proves that the PSO algorithm efficiently navigates the complex 700-dimensional parameter space (weights and biases) of the ELM network.

Stabilization Phase (Iterations 20-50): From iteration 20 onwards, the curve flattens, indicating the "Exploitation" phase. The particles fine-tune their positions around the global optimum. The stability of the curve in the final iterations suggests that the model has successfully converged and avoided

premature stagnation (getting trapped in local minima), which is a common issue in standard gradient-based training methods.

Ultimately, the algorithm identifies the optimal weight configuration at the end of the 50th iteration, ensuring the ELM network is fully optimized for the forecasting task.

5.3 Comparative Evaluation between Standard ELM and PSO–ELM

To comprehensively evaluate the **accuracy and stability** of the proposed hybrid model, both the conventional ELM and the PSO–ELM models were executed **20 independent times** under identical experimental conditions. The Mean Absolute Percentage Error (MAPE) obtained in each run is reported in table 5.1.

Table 5.1 Comparison of MAPE Results between Standard ELM and PSO-ELM across 20 Independent Runs

Run	Standard ELM (%MAPE)	PSO-ELM (%MAPE)
1	2.1723%	2.0000%
2	2.2765%	1.9800%
3	2.2014%	2.0100%
4	2.1256%	2.0500%
5	2.1805%	2.0349%
6	2.1808%	2.0159%
7	2.1727%	2.0394%
8	2.2662%	2.0657%
9	2.1742%	2.0130%
10	2.1427%	2.0130%
11	2.1484%	2.0674%
12	2.1193%	2.0430%
13	2.1439%	2.0059%
14	2.1563%	2.0363%
15	2.1959%	2.0061%
16	2.3058%	2.0060%
17	2.1565%	2.0273%
18	2.2628%	1.9626%

19	2.1554%	1.9683%
20	2.1675%	2.0031%

The results clearly show that the PSO–ELM model consistently achieves lower MAPE values than the Standard ELM in all experimental runs. This demonstrates that optimizing the hidden-layer parameters of ELM using PSO effectively reduces forecasting error and improves model reliability.

To further quantify the performance improvement, a statistical summary of the experimental results is presented in Table 5.2.

Table 5.2 Statistical performance evaluation and stability comparison

Method	Best (Min MAPE)	Worst (Max MAPE)	Average (Mean MAPE)	Std Dev (Stability)	Range (Max - Min)
Standard ELM	2.1193%	2.3058%	2.1852%	0.0510%	0.1865%
PSO-ELM	1.9626%	2.0674%	2.0174%	0.0279%	0.1048%

Several important observations can be made:

- **Accuracy:** PSO-ELM outperforms Standard ELM in every single run. Even the worst case of PSO-ELM (2.0674%) is better than the best case of Standard ELM (2.1193%).
- **Stability:** The Standard Deviation of PSO-ELM (0.0279%) is nearly half that of Standard ELM (0.0510%), proving that the hybrid model is much more reliable and consistent.
- **Reliability:** The narrower MAPE range of PSO–ELM confirms that the hybrid model effectively mitigates the randomness issue inherent in conventional ELM.

These results clearly demonstrate that integrating PSO into the ELM framework leads to substantial improvements in both forecasting accuracy and stability.

5.4 Discussion

Based on the experimental results, it can be concluded that the proposed **ELM–PSO model** significantly outperforms the conventional ELM approach for short-term electric load forecasting. The optimization of hidden-layer parameters using PSO not only enhances prediction accuracy but also ensures consistent performance across repeated experiments.

The strong convergence behavior, accurate load tracking, and improved statistical stability validate the effectiveness of the proposed hybrid approach and confirm its suitability for practical short-term load forecasting applications.

CHAPTER 6. CONCLUSION AND RECOMMENDATIONS

6.1 Conclusion

Short-term electrical load forecasting is a vital component in the operation and planning of power systems. This thesis addressed the specific challenge of improving forecasting stability and accuracy by proposing a hybrid model combining the **Extreme Learning Machine (ELM)** and **Particle Swarm Optimization (PSO)**. While ELM is renowned for its rapid learning speed and simple structure, it suffers from instability due to the random initialization of input weights and hidden biases. The integration of PSO was employed to optimally tune these parameters, thereby mitigating the stochastic nature of the standard ELM.

Based on the theoretical analysis and comprehensive experimental results presented in Chapter 5, the following conclusions can be drawn:

- **Superior Forecasting Accuracy:** The proposed PSO-ELM model demonstrated a consistent improvement in prediction accuracy over the standard ELM model.
 - The experimental results showed that the **Mean Absolute Percentage Error (MAPE)** was reduced from an average of **2.1852%** (Standard ELM) to **2.0174%** (PSO-ELM).
 - In the best-case scenario, the PSO-ELM model achieved a MAPE as low as **1.9626%**, proving its capability to capture the nonlinear and complex characteristics of the electrical load profile.
- **Enhanced Model Stability and Robustness:** The most significant contribution of this study lies in the improvement of model reliability.
 - Standard ELM models exhibit high variance in performance across different trials due to random initialization. The experiments revealed that the **Standard Deviation** of the MAPE for the Standard ELM was **0.0510%**.
 - The PSO-ELM model significantly minimized this fluctuation, achieving a standard deviation of only **0.0279%**. This **45% reduction in instability** confirms that the PSO algorithm successfully navigated the parameter space to find a stable and optimal network configuration, rather than relying on chance.
- **Feasibility for Practical Application:** Although the optimization process via PSO introduces additional computational time compared to the single step learning of standard ELM, the convergence analysis indicated that the model converges rapidly within the first 20 iterations. Thesidering the trade-off between a slight increase in training time and a substantial gain in

reliability, the PSO-ELM model is a highly viable solution for daily operational planning where accuracy and consistency are paramount.

6.2 Limitations

Despite the encouraging results, this study has several limitations that should be acknowledged:

- **Scope of Forecasting:** The model focuses exclusively on short-term load forecasting at the system level. The applicability of the proposed approach to medium-term or long-term forecasting scenarios, or to individual household levels, has not been investigated.
- **Input Feature Set:** The input features are primarily based on historical load patterns and dry-bulb temperature. Although this choice ensures model simplicity, it may restrict the ability of the model to fully capture external influences under extreme weather conditions or abnormal social events.
- **Simulation Environment:** The experiments were conducted in an offline simulation environment using historical data. Real-time deployment aspects, such as online learning capability and computational constraints in operational control centers, were not fully addressed in this study.

6.3 Recommendations and Future Work

To further enhance the applicability and precision of the forecasting system, several avenues for future research are recommended:

1. **Expansion of Input Features:** The current model relies heavily on historical load data and temperature. However, electricity demand is also sensitive to other meteorological and socio-economic factors. Future studies should be incorporated:
 - **Humidity and Heat Index:** As noted in related studies on load forecasting (e.g., Dang Quang Khoa, 2017 [9]), humidity significantly impacts human comfort and air conditioning usage.
 - **Special Events Calendar:** Encoding features for specific holidays, festivals, or major sporting events could help the model better anticipate abnormal load drops or spikes that statistical periodicity fails to capture.
2. **Investigation of Advanced Optimization Algorithms:** This thesis utilized Particle Swarm Optimization (PSO) due to its simplicity and global search capability. Future work could compare PSO with newer meta-heuristic algorithms such as the **Grey Wolf Optimizer (GWO)** or **Whale Optimization Algorithm (WOA)**, or evolutionary strategies like **Genetic Algorithms (GA)**, to potentially achieve faster convergence rates.

3. **Development of Deep Hybrid Models:** Recent benchmarks (Nguyen Thi Hoai Thu et al., 2024) [10] suggest that while Deep Learning models like LSTM generally require longer training times, they excel at capturing long-term temporal dependencies. A promising direction is to develop hybrid **LSTM-PSO** architecture. This would combine the deep memory capabilities of LSTM with the parameter optimization strengths of PSO, potentially offering a superior balance for medium-term forecasting horizons.
4. **Real-time Deployment and Online Learning:** Future development should focus on deploying the model as a web-based dashboard or integrating it into a SCADA system. Implementing an **Online ELM (OS-ELM)** mechanism would allow the model to update its weight continuously as new data streams in, ensuring the forecast remains accurate over time without the need for periodic retraining.

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